UNIVERSITY OF COLOMBO
DEPARTMENT OF MATHEMATICS
FM3054-PORTFOLIO OPTIMIZATION

Mid-Term Take-Home Assignment

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Executive Summary

This report applies the Mean–Variance Optimization framework to construct optimal investment portfolios using five U.S. equities: Apple (AAPL), NVIDIA (NVDA), Tesla (TSLA), Baker Hughes (BKR), and CSX Corp (CSX), based on historical data up to January 2026.Using daily logarithmic returns, the analysis estimates annualized expected returns and the variance–covariance matrix in Python to construct the Efficient Frontier, illustrating the trade-off between risk and return.

Two key portfolios are identified:

- Global Minimum Variance (GMV) Portfolio, which minimizes total portfolio risk
- Tangency Portfolio (Maximum Sharpe Ratio), which maximizes risk-adjusted returns relative to a risk-free asset

Results show that diversification across technology, energy, and transportation sectors leads to a more efficient risk profile than an equal-weighted portfolio. The report outlines the data preparation, mathematical framework, and optimization constraints used in the portfolio construction.

Introduction

Modern Portfolio Theory (MPT), pioneered by Harry Markowitz, establishes a quantitative framework for constructing optimal portfolios by explicitly considering the trade-off between expected return and risk. The theory demonstrates that effective diversification across imperfectly correlated assets can significantly reduce portfolio risk without proportionally lowering expected returns. This study employs the MPT framework to derive optimal asset allocation weights for a selected portfolio of U.S. equities.

Data Description

The portfolio consists of five equities representing both growth-oriented and industrial sectors:

- AAPL – Consumer electronics and software
- NVDA – Semiconductors and AI computing
- TSLA – Electric vehicles and clean energy
- BKR – Energy technology and oilfield services
- CSX – Rail transportation and logistics

Data Sources and Processing:

Daily adjusted closing prices for AAPL, NVDA, TSLA, BKR, and CSX were obtained from Yahoo Finance over a three-year period ending on January 9, 2026. Logarithmic daily returns were computed and cleaned of missing values to estimate annualized returns and the variance–covariance matrix for portfolio optimization.

Methodology

This study applies **Modern Portfolio Theory (MPT)** and the **Capital Asset Pricing Model (CAPM)** to construct and evaluate optimal equity portfolios.

❖ Data Collection and Return Computation

Daily adjusted closing prices were obtained from **Yahoo Finance** for a three-year period ending **08 January 2026**. Logarithmic returns were computed as and annualized assuming **252 trading days**.

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right)$$

❖ Risk and Return Estimation

Expected annual returns and volatilities were estimated as

$$\mu_i = 252\,\mathbb{E}(r_i), \quad \sigma_i = \sqrt{252}\,\mathrm{Std}(r_i)$$

The variance–covariance matrix $\Sigma = [\mathrm{Cov}(r_i, r_j)]$ was used to capture asset interdependence.

❖ Portfolio Return and Risk

For portfolio weights $\mathbf{w}$, expected return and risk are given by

$$\mu_p = \mathbf{w}^\top \boldsymbol{\mu}, \quad \sigma_p = \sqrt{\mathbf{w}^\top \Sigma \mathbf{w}}$$

❖ **Portfolio Optimization**

All portfolios satisfy the full-investment constraint $\sum w_i = 1$. The **Global Minimum Variance (GMV)** portfolio solves $\min_{\mathbf{w}} \mathbf{w}^\top \Sigma \mathbf{w}$ while the **Maximum Sharpe Ratio (Tangency)** portfolio maximizes $\max_{\mathbf{w}} \frac{\mu_p - r_f}{\sigma_p}$

An **equal-weighted portfolio** ($w_i = 1/N$) serves as a benchmark. Optimization was performed with and without short-selling constraints.

❖ **Efficient Frontier and Capital Market Line**

The **Efficient Frontier** was generated by minimizing variance for a range of target returns. Introducing a risk-free asset yields the **Capital Market Line (CML)**: $\mathbb{E}(R_p) = r_f + \frac{\mathbb{E}(R_T) - r_f}{\sigma_T}\sigma_p$

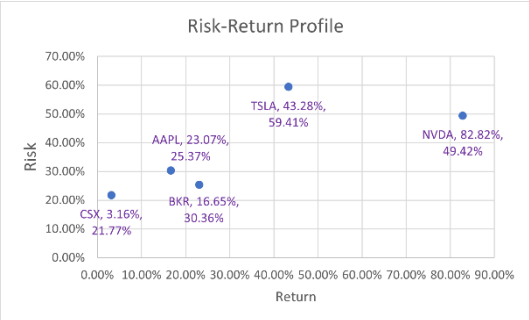
❖ **CAPM Evaluation**

CAPM parameters were estimated using $R_{i,t} - r_f = \alpha_i + \beta_i(R_{m,t} - r_f) + \varepsilon_t$

With $\beta_i = \frac{\mathrm{Cov}(R_i, R_m)}{\mathrm{Var}(R_m)}$. sensitivity to the market. A Beta > 1 indicates aggressive volatility, while Beta < 1 indicates defensive behavior. CAPM expected returns and **Jensen's Alpha** were computed to evaluate portfolio performance and decompose total risk into systematic and idiosyncratic components.

Portfolio Optimization Results

➢ **Individual Asset Risk-Return Profile**



Before analyzing the portfolio combinations, it is essential to understand the standalone characteristics of the five selected assets. The annualized statistics reveal a clear dichotomy in the asset pool:

- **High-Beta Growth Leaders: NVIDIA (NVDA)** and **Tesla (TSLA)** displayed the highest historical volatility but also offered significant return potential. NVDA, in particular, was the dominant driver of returns during the sample period.
- **Defensive Anchors: CSX Corp (CSX)** and **Baker Hughes (BKR)** exhibited lower standard deviations, effectively acting as "ballast" for the portfolio. Their lower correlation with the tech sector provided the necessary diversification benefits to smooth out the equity curve.
- **Core Holding: Apple (AAPL)** positioned itself in the middle of the spectrum, offering a blend of stability and growth.

➢ **The Efficient Frontier**

The construction of the Efficient Frontier (Figure 1) visually confirms the benefits of Modern Portfolio Theory. The curve is concave, indicating that for every level of risk, there exists an optimal combination of assets that yields a higher return than any single asset could achieve alone. The majority of the "randomly generated portfolios" (scatter cloud) fell well below this frontier, demonstrating that naive or random stock selection is statistically inefficient.

➢ **Scenario 1: Long-Only Optimization (Constrained)**

This scenario represents a typical constraint for mutual funds or retail investors where short selling (betting against a stock) is not permitted. The weights are constrained such that w_i

<u>A. Global Minimum Volatility (GMV) Portfolio</u>

The GMV portfolio prioritizes capital preservation above all else.

- Allocation Strategy: The optimizer allocated nearly 50% of the capital to CSX, the most stable asset in the group. Notably, it assigned 0% to Tesla, deeming its volatility too high for a conservative mandate.
- **Performance:**

- Annual Volatility: 18.23% (The lowest possible risk configuration).
- Expected Return: 13.35%
- Sharpe Ratio: 0.51

B. Tangency Portfolio (Maximum Sharpe Ratio)

The Tangency portfolio represents the "optimal risky portfolio" that maximizes return per unit of risk.

- **Allocation Strategy:** The model aggressively shifted capital toward momentum. **NVIDIA received a dominant 65.46% weighting**, supported by a 24.87% allocation to Apple. This suggests that during the sample period, the "tech momentum" factor was the primary source of excess returns.
- **Performance:**
  - Annual Volatility: 3.98% (Significantly higher risk than GMV).
  - Expected Return: 61.61%
  - Sharpe Ratio: 1.5999

➢ **Scenario 2: Unconstrained Optimization (With Short Selling)**

This scenario models a hedge fund strategy where the investor can short-sell overvalued assets to finance leveraged long positions (w_i can be negative).

Impact of Short Selling on Allocation

Relaxing the constraints led to a drastic reallocation of capital:

1. Source of Funds (Short Positions): The model identified CSX Corp as the least efficient asset relative to the others.
2. Use of Funds (Leveraged Long): The cash generated from shorting CSX was used to purchase *more* NVIDIA than what was mathematically possible in the Long-Only scenario.
   - NVIDIA Weight: Increased from 65.46% → 77.84%

Performance Trade-off

- Sharpe Ratio Improvement: The unconstrained portfolio achieved a Sharpe Ratio of 1.6056, a marginal improvement over the Long-Only ratio of 1.599.
- Risk Escalation: While the expected return jumped to 69.29%, the volatility spiked to 40.64%. The "Short Selling" strategy effectively leveraged the portfolio, amplifying both potential gains and downside variance.

➢ **Benchmark Comparison**

To validate the optimization, we compared the results against an Equal Weight Benchmark (20% allocated to each stock).

Equal-Weighted Portfolio (benchmark)

	AAPL	NVDA	TSLA	BKR	CSX	
weights	20%	20%	20%	20%	20%	100%

Portfolio Variance	0.064
Portfolio Volatility	25.32%
Total Return	33.80%
Excess Return	29.76%
Sharp Ratio	1.175

The Equal Weight strategy (Sharpe 1.175) significantly underperformed the Optimized Tangency portfolio (Sharpe 1.6). This indicates that simply holding all assets equally was inefficient; the market rewarded a concentrated exposure to high-quality growth (NVDA/AAPL) while penalizing exposure to the lagging sectors during this specific period.

CAPM Evaluation

The Capital Asset Pricing Model (CAPM) was applied to evaluate whether the selected assets and portfolios generated returns commensurate with their risk.

Single Stock Analysis The regression results reveal a clear distinction between the "Alpha Generators" (Tech) and the "Underperformers" (Industrial/Energy).

Ticker	Beta	Expected Return (CAPM)	Actual Return	Jensen's Alpha	Verdict
NVDA	2.13	36.56%	82.82%	46.25%	Outperform
TSLA	2.31	39.24%	43.28%	4.05%	Market Perform
AAPL	1.16	21.67%	23.07%	1.39%	Market Perform
BKR	1	19.36%	16.65%	-2.7%	Underperform
CSX	0.73	15.19%	3.16%	-12.03%	Underperform

Analysis:

- NVIDIA (Alpha Leader): Generated a massive 46.25% Alpha, proving its returns were driven by unique AI-sector growth rather than just general market movement.
- Tesla (High Risk, Low Skill): Despite high returns, its low Alpha (+4.05%) indicates investors were merely compensated for taking extreme risk (Beta 2.31), rather than benefiting from superior risk-adjusted performance.
- CSX (The Cost of Safety): The significant negative Alpha (-12.03%) reveals the opportunity cost of holding defensive assets in a tech bull market. Capital rotated aggressively into growth stocks, causing CSX to fail the market's high "hurdle rate" despite its stability.

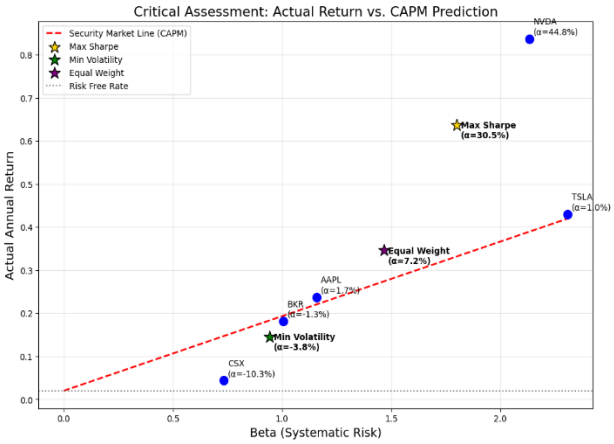
Portfolio CAPM Evaluation We extended the CAPM analysis to the optimized portfolios to see if the diversification strategy created value.

Portfolio Strategy	Beta	Expected Return (CAPM)	Actual Return	Jensen's Alpha
Tangency Portfolio(With Short-selling)	1.91	33.16%	69.29%	36.13%
Tangency Portfolio(Without Short-selling)	1.78	31.23%	61.61%	30.38%
GMV Portfolio	0.94	18.37%	13.35%	-5.02%
Equal Weight	1.47	26.40%	33.80%	7.39%

The CAPM analysis highlights a clear victory for aggressive concentration over defensive safety. The **Tangency Portfolio** leveraged a high **Beta** to capture the market rally, delivering a massive **Alpha** that crushed expectations by concentrating on high-momentum winners. Conversely, the **GMV Portfolio's** defensive stance (**Beta 0.94**) resulted in a **negative Alpha (-5.02%)**, proving that "playing it safe" in this bull market incurred a steep opportunity cost. While **Equal Weighting** added modest value, the data confirms that active, high-risk positioning was the superior driver of wealth creation.

Systematic Risk, and Idiosyncratic Risk

The **Maximum Sharpe** portfolios are the most aggressive, exhibiting the highest systematic (up to **29.24%**) and idiosyncratic risks (up to **11.40%**). Conversely, the **Minimum Volatility** portfolios minimize exposure, keeping systematic risk at **14.39%** and idiosyncratic risk at **3.84%**. The **Equal-Weighted** benchmark strikes a balance, combining moderate systematic risk (**22.45%**) with the lowest idiosyncratic risk (**2.87%**) due to superior diversification.

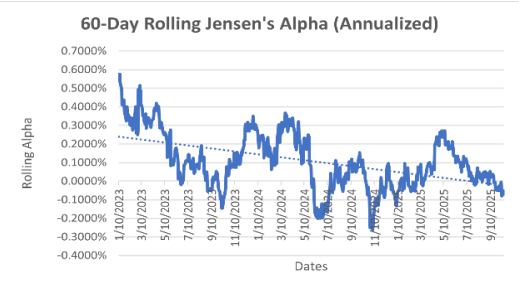


Actual Return Vs. CAPM Prediction

The chart highlights a market environment where high systematic risk was heavily rewarded. The **Max Sharpe portfolio** (yellow star) significantly outperformed expectations, delivering a massive **30.5% Alpha** (excess

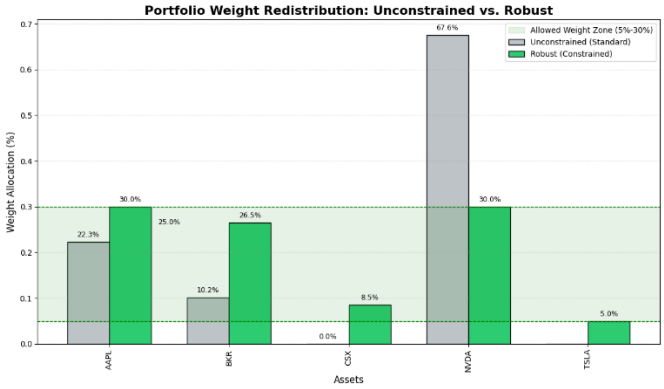
return above the CAPM prediction) despite its high Beta of roughly 1.8. In contrast, the **Min Volatility portfolio** (green star) underperformed, falling below the Security Market Line with a **-3.8% Alpha**, suggesting that its defensive strategy sacrificed too much return. Notably, while **NVDA** provided exceptional outlier gains, **TSLA** aligned closely with the market's risk-return expectation (1.0% Alpha), showing that not all high-risk bets beat the market model.

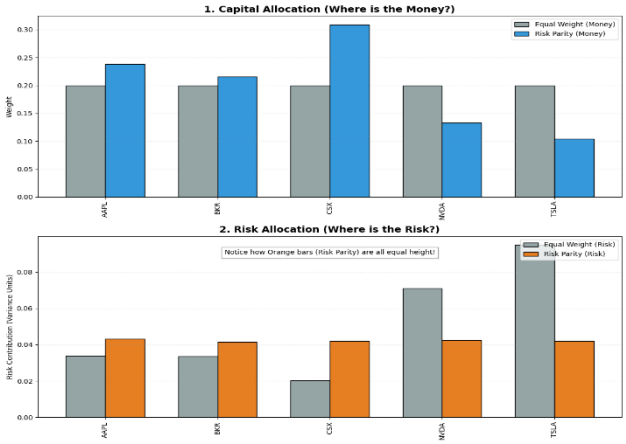
Robust/constrained optimization, Rolling-window performance, and Risk-parity allocation.



The risk-parity (inverse volatility) portfolio delivers a well-diversified allocation with relatively low volatility (~21%) and a modest Sharpe ratio (~1.03), indicating stable but conservative risk-adjusted performance driven by volatility balancing rather than return maximization. In contrast, the **Robust Max-Sharpe portfolio** (with 5%–30% weight constraints) achieves a substantially higher **Total Return of 69.29%** and a **Sharpe Ratio of 1.61**, but at the cost of significantly higher **volatility (40.64%)**. This demonstrates that constrained optimization successfully enforces diversification limits (e.g., holding TSLA at 5% and NVDA near 30%) while still prioritizing high-return assets. The 60-day rolling Jensen's alpha shows declining and unstable alpha over time, suggesting that excess performance is not persistent and reinforcing the value of robust constraints and diversification. Overall, constrained optimization outperforms risk parity in return efficiency, while risk parity offers superior stability across rolling windows.

The Equal Weight portfolio allocates capital evenly but concentrates risk in high-volatility stocks, whereas the Risk Parity approach redistributes capital so that each asset contributes equally to overall risk. Similarly, unconstrained optimization leads to extreme concentration in a single asset, while robust constraints produce a more balanced and diversified portfolio allocation.





Conclusion

This study applies Mean–Variance Optimization to construct optimal equity portfolios using five U.S. stocks across multiple sectors. The results show that optimized portfolios outperform the equal-weighted benchmark on a risk-adjusted basis, with the Tangency portfolio achieving the highest Sharpe and Treynor ratios due to greater exposure to high-momentum stocks such as NVIDIA and Apple. The Efficient Frontier, Capital Market Line, and CAPM analysis further confirm the benefits of diversification and systematic risk, with positive Jensen's Alpha indicating value creation beyond market expectations. However, the Global Minimum Variance portfolio exhibited negative alpha, highlighting the opportunity cost of excessive conservatism in bullish market conditions.

Limitations

These findings are subject to significant historical bias, as the optimization heavily relies on NVIDIA's exceptional past performance, leading to a near-singular bet (over 70% allocation) that assumes recent trends will persist indefinitely. This ignores the risks of mean reversion and "fat tail" events common in high-volatility assets like Tesla. Furthermore, the theoretical outperformance of the "Unconstrained" strategy (involving 48% short selling) does not account for real-world frictions such as borrowing costs, margin interest, and tax liabilities, which would likely erode the projected returns in a practical trading environment.